Statement of Significance

Two immiscible fluids mixed by the Rayleigh–Taylor instability form a turbulent layer that grows faster than one of miscible fluids, and recent simulations show that it grows faster still as surface tension increases, although surface tension slows the instability in linear theory. We show from the self-similar moment equations that surface tension can change the growth only through a correction that fades as the layer widens, that its direct force slows the growth, and that the acceleration comes from its effect on the flow structure. Simulations compared at equal layer width follow the predicted form.